

Roman Ibrahimov

CONTACT INFORMATION

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EDUCATION AND PROFESSIONAL DEVELOPMENT

University of California, Berkeley, CA, USA
M.S., Mechanical Engineering

Aug 2023–May 2025

Purdue University, West Lafayette, IN, USA
M.S., Aeronautics and Astronautics

Jan 2021–May 2023

- **IEEE RAS Summer School on Multi-Robot Systems**, CTU, Prague, Czechia, *1-5 Aug 2022*
Multi-UAV control, perception, localization, and planning
- **DroneCamp**, University of California, ANR, Monterey, California, USA, *27 June - 1 July, 2022*
UAV hardware & sensors, mission planning, flight skills, safety & regulations

ADA University, Baku, Azerbaijan
B.S. Summa Cum Laude, IT and Systems Engineering

Sep 2014–May 2018

- **ITMO University**, St. Petersburg, Russia, *Spring 2017*
Exchange Student, Control Systems and Robotics
- **Middle East Technical University**, Ankara, Turkey, *Spring 2016*
Exchange Student, Electrical and Electronics Engineering

RESEARCH INTERESTS

Planning & Control; State Estimation; Robot Learning; Contact-Rich Manipulation; Sim-to-Real Transfer;

PUBLICATIONS

R. Ibrahimov, S. Mozaffari, and A. Adel, “Contact-Rich Manipulation in Construction via Zero-Shot Learning: A Diffusion Policy Guided Adaptive Control,” *in preparation for Automation in Construction*, 2026.

R. Ibrahimov, T. Yang, and M. W. Müller, “Kalman Filter-Based Drift Detection and Mitigation of Visual-Inertial Odometry in UAVs,” *American Control Conference (ACC)*, 2025.

T. Yang, **R. Ibrahimov**, and M. W. Müller, “Towards Safe and Efficient Through-the-Canopy Autonomous Fruit Counting with UAVs,” *IEEE International Conference on Robotics and Automation (ICRA)*, 2025.

R. Ibrahimov and J. Heinen, “Multi-Robot Distributed Optimization for Exploration and Mapping of Unknown Environments using Bioinspired Tactile-Sensor,” *arXiv preprint*, 2025.

A. Behjat, **R. Ibrahimov**, A. Lenjani, A. Barket, K. Martinus, A. Maghareh, D. Whitaker, I. Bilonis, and S. Dyke, “A Computational Framework for the Evaluation of Resilience in Deep Space Habitat Systems,” *ASME 2022 International Design Engineering Technical Conferences & Computers and Information in Engineering Conference*, St. Louis, Missouri, USA, Aug. 14-17, 2022.

E. Tsykunov, A. Fedoseev, E. Dorzhieva, R. Agishev, **R. Ibrahimov**, D. Vasquez, L. Labazanova, and D. Tsetserukou, “DroneStick: Flying Joystick as a Novel Type of Interface,” *ACM SIGGRAPH Asia 2021 Emerging Technologies*, Virtual, Dec. 14-17, 2021.

E. Karmanova, V. Serpiva, S. Perminov, **R. Ibrahimov**, A. Fedoseev, and D. Tsetserukou, “SwarmPlay: A Swarm of Nano-Quadcopters Playing Tic-tac-toe Board Game against a Human,” *ACM SIGGRAPH 2021 Emerging Technologies*, Virtual, Aug. 9-13, 2021.

R. Ibrahimov, N. Zherdev, and D. Tsetserukou, “DroneLight: Drone Draws in the Air using Long Exposure Light Painting and ML,” *29th IEEE International Conference on Robot & Human Interactive Communication (IEEE RO-MAN 2020)*, Naples, Italy, Aug. 31-Sept. 4, 2020.

E. Tsykunov, R. Agishev, **R. Ibrahimov**, T. Moriyama, L. Labazanova, H. Kajimoto, and D. Tsetserukou, “SwarmCloak: Landing of Two Micro-Quadrotors on Human Hands Using Wearable Tactile Interface Driven by Light Intensity,” *2020 IEEE Haptics Symposium*, Washington DC, USA, March 28-31, 2020.

R. Ibrahimov, E.Tsykunov, V. Shirokun, A. Somov, and D. Tsetserukou, “DronePick: Object Picking and Delivery Teleoperation with a Drone Controlled by a Tactile Wearable,” 28th IEEE International Conference on Robot & Human Interactive Communication (IEEE RO-MAN 2019), New Delhi, India, 2019.

E. Tsykunov*, **R. Ibrahimov***, D. Vasquez, D. Tsetserukou, “SlingDrone: System for Navigation and Interaction with the Environment Using a Single Drone and VR,” 25th ACM Symposium on Virtual Reality Software and Technology (VRST 2019), Sydney, Australia, 2019.

*- authors contributed equally to the paper.

E. Tsykunov, R. Agishev, **R. Ibrahimov**, L. Labazanova, T. Moriyama, H. Kajimoto, D. Tsetserukou, “Swarm-Cloak: Landing of a Swarm of Nano-Quadrotors on Human Arms,” Int. Conf. on Computer Graphics and Interactive Technologies (ACM SIGGRAPH Asia 2019), Emerging Technologies, Brisbane, Australia, 2019.

E.Tsykunov, R. Agishev, **R. Ibrahimov**, L. Labazanova, A. Tleugazy, and D. Tsetserukou, “SwarmTouch: Guiding Swarm of Nano-Quadrotors with Impedance Control using Wearable Tactile Interface,” IEEE Transactions on Haptics, 2019.

G. Yashin, D. Trinitatova, R. Agishev, **R. Ibrahimov**, and D. Tsetserukou, “AeroVR: Virtual Reality Teleoperation System for the UAV Robotic Manipulator,” 19th IEEE International Conference on Advanced Robotics (ICAR 2019), Belo Horizonte, Brazil, 2019.

PROJECT EXPERIENCE

Contact-Rich Robotic Manipulation for Construction, Princeton University July 2025–present

- Developing a contact-rich manipulation framework for automated assembly tasks using a UR20 robot.
- Generating large-scale synthetic demonstration datasets in IsaacSim for robust sim-to-real transfer.
- Training and evaluating diffusion policy augmented with adaptive control for zero-shot real-world performance.

Autonomous Drone Inspection of Caltrans Bridges, UC Berkeley Aug 2023–May 2025

- Implemented visual-inertial odometry with OpenVINS, integrating IMU and stereo camera.
- Configured and tested hardware, including RealSense D455 for depth imaging, Qualcomm RB5 for onboard computing and Pixracer R15 for flight control.
- Managed system communication via the Robot Operating System (ROS).

An inspection robot for Duke Energy Power Lines, Purdue University May 2022–Aug 2023

- Implemented Model Predictive Control (MPC) for trajectory tracking along power lines.
- Developed line detection algorithms using OpenCV for enhanced visual recognition.
- Modeled and simulated robot dynamics.
- Configured and tested Argo rover for optimal field deployment and reliability.

Resilient Extraterrestrial Habitats, NASA RETH Institute, Purdue University May 2021–Aug 2023

- Designed a control-theoretic autonomy framework to support resilient design and operation.
- Built automated active learning framework with robots and humans-in-the-loop.
- Developed methods for detection and diagnosis of anticipated and unanticipated faults.
- Contributed to establishing SmartHabs with autonomous abilities to sense, anticipate and respond.

Bio-inspired nano-quadcopter for map building, Purdue University Sep 2021–May 2022

- Developed an API on FreeRTOS to read sensor reading on the quadcopter.
- Transmitted sensor data to the ROS base station via radio communication.
- Collected and processed point cloud data to construct detailed environmental maps.
- Applied machine learning techniques to predict the map of the environment.

Human-Drone Interaction through a Tactile Wearable, Skoltech Oct 2018–May 2020

- Implemented human-drone communication using impedance control and vibrotactile feedback.
- Designed a tactile wearable equipped with eccentric rotating mass (ERM) motors for intuitive interaction.
- Created a Virtual Reality (VR) application in C# to support teleoperation and aerial manipulation tasks.
- Developed remote object manipulation techniques using drones.

Balloon Satellite for Testing Solar Cells in High Altitude, Skoltech Sep 2018–Jan 2019

- Applied Model-based Systems Engineering principles to ensure mission success.
- Constructed a CubeSat featuring an onboard controller, solar cells, storage devices, and GPS tracking.
- Gathered and analyzed current, voltage, and temperature (CVT) data from solar cell tests.
- Retrieved payload with no damage after landing from 35km maximum altitude.

WORK EXPERIENCE

Gas Leak Detecting Mobile Robot for NICA Collider, JINR

Summer 2018

- Designed autonomous navigation system around elliptical collider.
 - Built a mobile robot with an on-board temperature camera.
 - Developed Computer Vision (CV) algorithm based on Python for detection nitrogen gas leak from the collider.
 - Came up with computer-based user interface for remote monitoring.
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Robotics Research Associate, Princeton University, NJ, USA

Jul 2025–present

- Develop contact-rich robotic manipulation strategies for construction assembly using UR20 manipulators.
- Generate large-scale synthetic datasets in Isaac Sim for sim-to-real transfer.
- Implement ROS 2 control pipelines with force/torque feedback for contact-rich manipulation.
- Collaborate with faculty and students on experimental design, system integration, and publications.

Research Co-Op, MaxInsights, Inc., San Jose, CA, USA

Jan 2025–May 2025

- Built a teleoperation pipeline integrating DexCap motion capture with Rokoko gloves on a Realman manipulator for dexterous manipulation data collection.
- Developed real-time data streaming and coordinate-frame calibration between the glove, motion capture system, and robot end-effector.
- Contributed to data collection workflows for training embodied AI policies on dexterous manipulation tasks.
- Demonstrated the full pipeline live at IEEE ICRA 2025.

Graduate Student Instructor, UC Berkeley, CA, USA

Jan 2025–May 2025

- ENGIN07 Computer Programming for Scientists and Engineers.
- Led coding labs and exercises, supporting course delivery.
- Conducted hands-on sessions to apply concepts in programming tasks.
- Held office hours to answer questions and assist with assignments.
- Designed and graded coding assignments, providing feedback on student work.

Graduate Research Assistant, UC Berkeley, CA, USA

Aug 2023–May 2025

- Designed and tested control algorithms for UAV stability and obstacle avoidance in complex environments.
- Optimized data processing workflows, enhancing efficiency and accuracy for real-time inspection feedback.
- Collaborated with interdisciplinary teams to troubleshoot hardware and software challenges.
- Presented research findings within lab meetings and contributed to technical publications on UAV autonomy.

Graduate Research Assistant, NASA RETH Institute, IN, USA

May 2021–Aug 2023

- Developed real-time fault detection algorithms and situational awareness systems to enhance safety and reliability in extreme environments.
- Designed and configured hardware systems, including specialized sensors, actuators, and computational units, to support autonomous operation in space habitats.
- Conducted simulations and tests to validate habitat resilience against environmental and operational disruptions.
- Collaborated with interdisciplinary teams to integrate control-theoretic approaches into habitat design, advancing NASA's SmartHab initiative with enhanced autonomous capabilities.

Teaching Assistant, Purdue University

Jan 2021–May 2022

- CNIT 155 Introduction to Software Development Concepts (in Python).
- Supported course delivery by leading coding labs and exercises, guiding students in foundational Python programming and problem-solving.
- Conducted hands-on lab sessions, helping students implement concepts through coding exercises and real-world applications.
- Held office hours to address individual student questions, clarify complex topics, and provide assignment assistance.
- Created and evaluated coding assignments, offering constructive feedback to enhance students' programming skills and understanding of software development principles.

Intern, Universal Robots, Moscow, Russia

Summer 2019

- Assisted in research and development for Human-Robot Collaboration (HRC) applications involving quadcopters to enhance interaction and safety.
- Participated in designing and testing control strategies to improve quadcopter responsiveness and collaboration capabilities.

- Supported hardware configuration and troubleshooting to ensure seamless integration of HRC functionalities.
 - Documented findings and presented insights to the team, contributing to advancements in HRC applications.
- Intern, Azercosmos OJSCo, Baku, Azerbaijan** *Summer 2017*
- Supported the Ground Control Department in monitoring and maintaining satellite communication systems.
 - Assisted with network configuration and troubleshooting to ensure stable connectivity and optimal system performance.
 - Gained experience in analyzing and managing data flow within ground control networks.
 - Documented operational procedures and reported on system performance, contributing to improved workflow efficiency.
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HONORS AND AWARDS

American Control Conference Full Travel Grant \$2000	<i>April 2025</i>
Purdue University Poster Symposium Best Poster Award \$750	<i>April 2022</i>
Skolkovo Foundation Full Graduate Scholarship	<i>2018-2020</i>
ACM SIGGRAPH Asia Best Demonstration Award	<i>November 2019</i>
Skolkovo Presidential Scholarship	<i>November 2019</i>
Skolkovo Presidential Travel Award, \$5000	<i>November 2019</i>
Skolkovo Presidential Travel Award, \$4500	<i>November 2019</i>
President's List of Honor, ADA University	<i>2014-2017</i>
Erasmus+ Exchange Scholarship, METU	<i>Spring 2016</i>
FameLab World Finalist, Cheltenham, UK	<i>June 2015</i>
1st place, Microsoft ImagineCup Competition, Baku, Azerbaijan	<i>May 2015</i>
4th place, International Rudolf Ortway Competition in Physics, Hungary	<i>Dec 2014</i>

MENTORING EXPERIENCE

Princeton University:
- Joshua Song, BS in Electrical and Computer Engineering
- Moyu Yamaguchi, BS in Civil and Environmental Engineering
UC Berkeley:
- Jannik Matthias Heinen, MEng in Mechanical Engineering (PhD student at Aalto University, Finland)
- Milan Rosic, MEng in Aerospace Engineering (safety engineer at Lucid Motors)
- Yuxuan Peng, MEng in Mechanical Engineering
- Ruonan Yang, MEng in Mechanical Engineering
Purdue University:
- Anderson Xu, BS in Aero/Astro (PhD student at Purdue University)
- Benjamin Krugman, BS in Mechanical Engineering
- Fahad Tajiki, BS in Electrical Engineering (firmware engineer at Saab, Inc.)

REVIEWER FOR CONFERENCE AND JOURNALS

IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) 2026, IEEE Robotics and Automation Letters (RA-L) 2024, RA-L 2023, RA-L 2022, IEEE International Conference on Robotics and Automation (ICRA) 2024, ICRA 2023, ICRA 2021, ICRA 2020; ACM Conference on Human Factors in Computing Systems (CHI) 2020; Virtual Reality & Intelligent Hardware Journal 2020;

TECHNICAL SKILLS

Programming: C/C++, Java, Python, MATLAB/Simulink, LabVIEW, L^AT_EX

Robotics: Robot Operating System (ROS), ROS2, Gazebo, NVIDIA Isaac Sim, RViz, Unity

MCUs: Qualcomm RB5, myRIO, Arduino, Libelium

Mechanical: CAD (SolidWorks), 3D Printing, soldering, laser/plasma cutting

